Reflection Journal Report for Lab 10

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The purpose of this report is for each student to reflect on their learning journey for ITAI 1371 Machine Learning (ML) Lab 10 This Reflection Journal Report consists of separate parts for each student in the group. Each student in the group was responsible for their own content and report formatting in each of the sections below.

This reflection journal entry is for Lab 10 Unsupervised Learning and was prepared by Eric Roberson. I am not back up to par yet, so this reflection journal entry will not be as fastidious as usual.

Understanding Insights & New Breakthroughs

The key takeaway from this module is an algorithm can find structure in data without being given any labeled training data. K-Means does this by placing centroids, assigning each point to the nearest one, and moving the centroids until they stop shifting. I also learned that PCA does not just delete features. Rather, It builds new ones (principal components) that capture the most variation, so I can shrink the data while keeping most of the information. The "best camera angle" analogy from the slides made that idea click for me. Pun intended!

Analyzing the Problem/Problem Solving Strategies

In the lab, I generated synthetic data with make_blobs, then used the Elbow Method to pick "k" by plotting inertia for k = 1 to 10. The curve bent at k = 4, so I ran K-Means with four clusters and colored the points by assignment. Since the data had no labels to check against, I had to rely on the elbow plot and the visual result to judge whether the clusters made sense. I also hit a syntax error early on because pasted code lost its line breaks, which taught me to read the error output closely before assuming the logic was wrong.

Real-World Applications

The slides describe K-Means used for customer segmentation, image segmentation, and anomaly detection. Segmentation lets a retailer group

shoppers into types like bargain hunters and premium buyers without defining those groups in advance. Anomaly detection stood out most because points that fit no cluster can signal fraud or intrusions, which connects directly to my work in technical risk and controls. PCA is also useful in practice for visualizing data with dozens of features.

Connecting New & Prior Knowledge

Supervised learning has both X and y: the input features and the labeled answers from which the model learns. Unsupervised learning has only X, so the algorithm has to find patterns on its own. In my mid-term using the Netflix dataset, I engineered a kid_safe target from the rating column, which is a supervised setup because every movie has a label to predict. With unsupervised learning, one could instead cluster the movies by their features and see whether natural groups emerge without ever creating that label. If we will be using the same Netflix dataset for the final, this is what we will do.

Emerging Questions

How do I choose k when the elbow isn't a clear bend (like on messy blob of real-world data)? How should I interpret what each principal component actually represents, since they are combinations of the original features? I'm also curious whether cluster assignments could be used as input features for a supervised model, and how that would affect my Netflix finals work if that's the dataset we will use?

Changing Approach & Critical Thinking

Before this module, I knew about unsupervised learning but was unsure how it worked exactly. It is still not completely clear to me, but at least I have a fundamental understanding of unsupervised learning. In hindsight, I can explore data first with clustering or PCA to understand its structure before deciding on a modeling approach. This would save considerable time and effort with training an test data. I also learned to be more skeptical of results, since K-Means is sensitive to starting points and assumes roughly round clusters. A clean-looking plot doesn't guarantee the groupings are meaningful. Next time I'd test more than one k- value and compare results before committing to one.

This reflection journal entry is for Lab 10 Unsupervised Learning and was prepared by Irene “Alex” Ramos:

This report is meant to be a reflection on the process involved in the study of ITAI 1378 Machine Learning Lab 10. This lab involves studying different concepts in unsupervised learning, namely the application of K-means clustering using the elbow method and dimensional reduction by use of PCA.

Understanding Insights & New Breakthroughs

An important achievement in this lab was the realization that unsupervised learning is done completely without target labels (y), hence the need for algorithms that are capable of finding hidden patterns and data structure using only the input features (X). The process of how K-Means updates its cluster centers and the use of the Elbow Method to evaluate the inertia for k values from 1 to 10 provided a method to determine the best number of clusters.

Analyzing the Problem / Problem Solving Strategies

As part of the process of implementing the task, artificial data of customer expenses were created via the make_blobs function, after which K-Means algorithms (n_clusters=4, random_state=42, n_init=10) were created and fitted. Centers of the clusters were found for plotting their location on a chart. In addition, PCA transformation of the 4-dimensional Iris dataset was performed to show that transforming the dataset into orthogonal principal components leads to lower dimensionality with preservation of most of the variance.

Real-World Applications

The discovery of natural groupings and dimension reduction will have clear practical consequences in system designs when applied in various areas like market segmentation, anomaly detection, and compression. Using principal component analysis (PCA), one is able to visualize the dataset while maintaining more than 97% of its variance in a small number of dimensions.

Connecting New & Prior Knowledge

This lab experiment combines the basic supervised process with the more exploratory unsupervised approach. Understanding the way in which dimensionality reduction

captures variation in the data emphasizes the importance of proper feature scaling before feeding data into a classifier or clustering algorithm.

Emerging Questions

As I looked into the explained variance of the principal components and the inertia curve of the same, the problem cropped up: how can the data scientist find out the right number of clusters as well as the cut-off value of the components while dealing with the real-world datasets which are high-dimensional and noisy?

Changing Approach & Critical Thinking

Moving forward, exploratory and unsupervised methods like PCA and clustering will be a part of the data analysis process at the beginning of any project before performing any kind of predictive modeling. It is vital to have an insight into structural variance to develop a clean data pipeline.

This reflection journal entry is for Lab 10 Unsupervised Learning and was prepared by Carmen Esono

Exploring the nuances of unsupervised learning in this lab provided a clear foundational shift from supervised workflows by focusing entirely on discovering hidden structures within unlabeled data rather than predicting pre-defined targets. Through K-Means clustering, the lab demonstrated how algorithms iteratively organize data points by randomly placing centroids, assigning nearest points, and updating positions until convergence. Working with synthetic customer data illustrated how clustering groups observations together based on proximity, making it an essential technique for customer segmentation.

critical highlight of the clustering workflow was utilizing the Elbow Method to determine the optimal cluster count (k=4) rather than arbitrarily guessing or maximizing k. By tracking inertia—the sum of squared distances of samples to their closest cluster center—the plot revealed a distinct "elbow" point where the rate of inertia reduction slows drastically. This step reinforced why choosing the largest possible k must be avoided: while it reduces inertia down toward zero, it results in severe overfitting where every individual data point forms its own isolated cluster instead of capturing meaningful, generalized patterns.

Transitioning into Principal Component Analysis (PCA) introduced the mechanics of dimensionality reduction, demonstrating how high-dimensional feature spaces can be compressed into orthogonal components ordered by the amount of variance they explain. Applying PCA to the 4-dimensional Iris dataset revealed that the first two principal components captured over 97% of the total variance, reflecting strong inter-correlations among the original

features. This high percentage proved that complex datasets can be successfully simplified and visualized in two dimensions with minimal information loss, bridging theoretical concepts with practical data science execution.

This reflection journal entry is for Lab 10 Unsupervised Learning and was prepared by Xiomara Torres:

The focus on this lab was to understand and run through an Unsupervised Learning exercise when training a machine learning model. For this lab, we were asked to discover patterned hidden data using k-means and Principal Component Analysis (PCA). The purpose of this reflection is to explain reasoning and understanding gained from this assignment.

This week's presentation focused on unsupervised learning and the two subsets of it, K-means and PCA. We started off the lab by importing the proper libraries, matplotl qqib, and sklearn.datasets. We then generated synthetic data and asked the program to visualize it for us. What followed was a pretty and organized graph that showed four clusters of data points. This was a proper visualization of the k-means learning model. I did not run into any issues with this part of the lab

Task 1 of this lab focused on finding the optimal number of clusters using the elbow method which was k = 4. I ran into some syntax errors in this part of the lab, but it was easily corrected with indentation. We then moved on to task 2 which plugged in that k = 4 value and created a scatterplot showing the clusters using that k = 4 value. Th graph returned with the same clusters as in the beginning, the only difference was now a small red x was in the middle of each cluster.

Lastly, we used the iris dataset to transform the data into a new system of principal components. For this lab i had to include my own code import matplotlib and import seaborn because the plot was not displaying properly at the end.

I felt confident when running this lab independently as the concepts have become more familiar. I was able to troubleshoot the code and understood the data.

This reflection journal entry is for Lab 10 Unsupervised Learning and was prepared by

[Yuanyuan Kou]:

In my earlier machine learning labs, I was used to asking what the model was trying to predict. In this lab, the question was different. K-Means looked for groups in the customer data, and PCA looked for patterns in the flower measurements. The Iris species names made the PCA graph easier to read, but PCA did not use those names when it learned from the data. That helped me understand what unsupervised means in practice.

The elbow plot helped me understand why I should look at the whole graph before choosing a number. Inertia gets smaller as we add more clusters, so it would be easy to pick the largest number of clusters. But after four clusters, the line became much flatter. Adding more groups made only a small difference. This helped me see why four clusters were a reasonable choice for this dataset.

PCA was harder for me to understand because the two axes in the new graph were different from the original flower measurements. Each axis combined information from the original measurements. The first two components kept more than 95% of the variation, so I could look at most of the variation in one graph. I also learned that explained variance and accuracy answer different questions. Explained variance tells me how much of the variation is kept. Accuracy would tell me how often a model predicted the correct species.

This lab made me pay more attention to what goes into each step and what comes out. One thing I would like to try next is scaling the flower measurements before running PCA. Then I could compare the two graphs and see how much that change affects what I observe.